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Supplementary Materials for

Endangered species recovery: A resource allocation problem

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Recovery Explorer Tool

We developed the Recovery Explorer tool to allow decision makers to explore how different funding allocation scenarios compare to one another. With input data on the relative attributes of possible choices, the tool applies a ranking algorithm to prioritize choices according to manager objectives, and then allows for the visualization and comparison of the implication of manager choices. Below we describe the process that the tool uses to rank data based on the relative benefit, costs and weights and success of different projects, and how we compiled the data used

to generate the figure that demonstrates the utility of a cost-effectiveness approach. A prototype version of the tool is available on the web at https://shiny.sesync.org/apps/RecoveryExplorer

Approach to prioritization

To demonstrate proof of concept in applying our tool, we used the following algorithm to prioritize allocation of funding to recovery plans according to their cost effectiveness (*1*, *2*):

$$E_i = \frac{W_i \times B_i \times S_i}{C_i} \quad \text{Eq. 1}$$

Here, project efficiency (E_i) depends on the benefit (B_i) that will result from funding and completing all actions for recovery plan *i*, the probability that the actions within the plan will be successful (S_i), any predetermined weights (W_i) for species *i* that describe additional values about which society cares, and the cost (C_i) of completing all actions listed within the recovery plan (corrected to account for inflation). The ranked list of project efficiencies can be used by a decision maker to identify the best plans to fund within a given fixed budget, where plans are prioritized in order of highest to lowest project efficiency, and projects are selected for action in order of priority until the budget is exhausted. This approach represents an extension of the "knapsack problem", which has been previously applied to species prioritization (*1*, *2*). We develop these early applications of the knapsack approach by working closely with U.S. Fish and Wildlife Service (USFWS) to use information found in species recovery plans to estimate the necessary parameters. Our intent was to explore how this approach (knapsack problem as a

prioritization tool) could be used to help inform decisions for the most species and within the most complex agency decision making structure to date.

We designed the recovery explorer tool to allow managers to consider several different possible management objectives. The statutory language suggests two central objectives (Fig. 2). First, the definition of an endangered species ("in danger of extinction throughout all or a significant portion of its range", 16 USC §1532(6)) identifies a primary concern of the ESA—the extinction of species. Thus, one potential objective for recovery funding is *to minimize the number of species that become extinct*. Second, the definition of conservation cited above focuses on taking action that results in a species no longer needing the protections of the ESA. Thus, a second potential objective supported by the tool is *to maximize the number of recovered species* (species removed from listing because their threats have been eliminated or mitigated). We also include scenarios for managers to explore that include abilities to set objectives that include criteria on taxonomic or regional inclusion.

As written in Eq. 1, and as implemented in the Recovery Explorer tool, the algorithm implies that species are independent, such that investment in any one species does not change the efficacy of investment in any other species. This is based on the assumption inherent within recovery plans, but known to be an oversimplification by managers, that all of the actions listed in the plan are necessary for the given species recovery. It is true that some of these actions may benefit other listed species, but we are also assuming that multispecies plans have been developed to account for these overlaps.

This is not, however, a limitation of the algorithm: Eq. 1 can be modified to account for how the probability of success (S_i) or the cost (C_i) for a particular recovery plan may depend upon which other plans are also funded. Previous work by McDonald-Madden et al. (*3*) demonstrates a more

complex approach that includes species interactions, and Brazill-Boast et al. (*4*) describe how to include cost sharing for species and explicitly account for the situation where managing for one species positively influences another species.

We chose not to include these complexities because the intent of the recovery explorer tool is to allow managers to explore the implications of changes to national level funding allocation policy rather than dictate spending on specific species.

Data for US species

To explore the potential benefits prioritizing recovery spending based on cost efficiency (Eq 1, and Fig 1 main text), and to demonstrate how the Recovery Explorer Tool can be used to visualize the outcomes of different scenarios, we needed to identify comparable data across recovery plans. We obtained data on the 598 active recovery plans contained in the Environmental Conservation Online System (ECOS) database (http://ecos.fws.gov/ecp/) by a core database query on January 10th, 2017. These data are available to the public via the Recovery Online Activity Reporting (ROAR) module of the database. About half of the data in the database comes directly from the published recovery plans (draft and final). This includes items like the plan title, species included in the plan, the total time to and cost of recovery, and the implementation schedule (actions, estimated costs per action, etc.). Other items in ROAR are not sourced from the published recovery plan, but are entered by USFWS or NOAA biologists on a yearly basis. These include the action status and recovery action start and completion dates. Active status recovery plans are the current version of each plan on the query date. We based our analysis on active recovery plans, because it is the best available information. However, many of these plans have not been updated recently, many listed species do not yet have recovery plans,

and older plans tend to not account for climate change effects. In addition, cost data are inconsistent across plans; our tool allows users to address uncertainty of cost estimates.

The raw data include the individual actions specified for each year and species covered by a recovery plan. We then matched species in the dataset to a current Recovery Priority Number (RPN) that provides an 18-category numeric ranking system for listed species based on recovery potential, degree of threat, taxonomic uniqueness, and degree of conflict (*5*).

Many of the recovery plans concern only a single listed entity, but there were 105 plans that included multiple species. For this analysis, we made the simplifying assumption that for multi-species recovery plans each action benefited all species in the plan, and that each species requires all the actions for recovery. In 27 cases where species within multi-species recovery plans did not have RPNs, we assigned values for benefit, success, and taxonomic weighting (explained below) that were the mean value of those for the other species in the multispecies plan. In these 27 cases, we were also missing information on listing status (Endangered vs. Threatened) so we assigned the most common status from the multi-species plan in which they occurred. In light of this artifact of the database structure, our results may overestimate the benefit of implementing these 27 multispecies plans because these species often were ones that are not yet listed.

We included only those plans for species listed as “Threatened” or “Endangered,” and excluded those without detailed information on the costs of actions or RPNs. The final dataset had 569 recovery plans, representing 1315 “species” (taxonomically distinct units).

To calculate the project efficiency of each plan using Eq. 1, we needed to determine the benefit (*B*), cost (*C*), and feasibility (*S*) of successfully implementing each plan, and assign any weightings (*W*) to reflect relative preference among species.

Estimating Benefit (B)

The benefit of fully funding a recovery plan for a given species is an expression of the multiple objectives that are fundamental to the allocation decision. Thus, the benefit value, B_i, for a particular recovery plan could be an aggregate measure that reflected such things as the value of recovery, the value of preventing extinction, the degree to which recovery was in conflict with economic activity, or other considerations. Without presuming that we have captured the statutory and societal values correctly, and for the purposes of demonstration, we implemented the following measure of benefit. The benefit of funding recovery plan *i* is the difference in the probability that the species would be recovered in 50 years if the plan was (P_{i1}) or was not (P_{i0}) implemented (that is, $B_i = P_{i1} - P_{i0}$). This benefit metric was one of many that could be chosen (*6*), but we used this one for the prototype analysis because it is conceptually intuitive. The timeframe is an important element of the objective; we chose 50 years, because it captures a long-term desire for viability of species without requiring very long-range forecasts. For recovery plans that contained multiple species, the benefit of funding each recovery plan *i* was considered to be the sum of the benefit for all the *n* species within the plan:

$$B_i = \sum_{j=1}^{n} (P_{i1j} - P_{i0j}) \qquad \text{Eq. 2}$$

This probability of recovery is related to the probability of extinction, and thus may be calculated from stochastic population models (*7*). However, because population viability models are not available for all species, we estimated the probabilities from information embedded in the RPNs which are assigned to each species by the USFWS. RPNs are based on estimates of recovery potential, degree of threat, and taxonomic distinctiveness (*5*). We used the estimated recovery potential to represent the probability of the species being secure in 50 years if the plan was

implemented (P_{i1}). The assigned recovery potential relates to knowledge of the biological and ecological factors limiting the species' population, how well understood, pervasive, and easily alleviated the threats are, and the intensity and probability of success of required management. For the RPN, the recovery potential of a listed species could be high or low. Noting again that these are placeholder values for the purposes of demonstration, we assigned a P_{i1} value of 1 to species with high recoverability and a value of 0.5 to those with low recoverability.

While we have formulated the problem to recover as many species as possible by a fixed time, other approaches may seek to maximize the net sum of species lifetimes (*8*). We opted for a simple objective for demonstration purposes, and to invite deliberation by the decision-making agencies. Ultimately, specification of the benefit metric is a policy judgment, not a scientific one.

We used the estimated degree of threat to represent the probability of the species being secure in 50 years if the plan was not implemented (P_{i0}). This value relates to the magnitude and immediacy of threats; in the RPN system, the value can be either high, medium, or low. For the purposes of demonstration, we assigned P_{i0} values of 0.01, 0.34, and 0.67 to species with a high, medium, or low degree of threats, respectively.

Our implicit assumptions are (i) that a species with a high recovery potential will benefit from management, and thus have a higher probability of being secure in 50 years than a species with a low recovery potential; and (ii) that a species with a high degree of threat is less likely to be secure in 50 years without management than a species with a low degree of threat. This approach meant that the six plans with low threat and low recovery potential resulted in a negative benefit (-0.17). We rescaled these benefit values to 0.01 for analyses because our sensitivity test assumed a Beta distribution for this variable. The final benefit values are shown in Table 1.

Estimating Success (S)

The biodiversity benefit that is expected to be achieved by implementing a recovery plan only manifests if the recovery actions perform as expected. The probability of success term (*S*) allows variation in expected success of recovery actions to be explicitly incorporated into the prioritization.

For this analysis, the probability of success for each recovery plan was set to 1 to correspond with the assumption of the recovery planning process that, if all of the actions were performed as prescribed in the plan, the species would be recovered to the point where it could be removed from the Federal List of Endangered and Threatened Wildlife and Plants.

We acknowledge that this is a strong assumption and not one that is universally held by those who implement the plans. These probabilities of success could be obtained through an expert elicitation process if desired.

Estimating Weights (W)

Weighting values can be incorporated into the ranking process to capture information on values important to the objectives of society or the managing agency. Thus, the weights (W_i) interact with the benefit values (B_i) to reflect the fundamental objectives of recovery for species *i*, relative to other species. In this prototype analysis, for the purposes of demonstration, we incorporated a weight for taxonomic distinctiveness, based on the taxonomic distinctiveness component of the RPN. A monotypic genus was assigned a score of 1, species were assigned 0.67, and subspecies were 0.33. For multi-species recovery plans, we used the average of the individual scores for each species in the plan.

Estimating Costs (C)

The cost of recovery was calculated as the total cost of implementing all the actions prescribed by all parties by a plan. For this analysis, we needed to be able to consider the costs of actions so we estimated relative costs across plans based on information included in each plan. The recovery plan dataset included information for each action on the first year that the action occurred, detailed costs for a number of years of funding (often five, but it differed across plans), and the duration of each action. We estimated the total cost of each action by summing the average inflation-corrected costs across (i) the years they were detailed, plus (ii) an extrapolation of the last listed cost for the remaining duration of last action listed (truncated to 50 years if the recorded duration exceeded that), plus (iii) an extrapolation of the first listed cost to the first listed year of action if the first detailed cost occurred after that year (this could happen for plan updates). In cases where action duration was not recorded, we used listed costs only. We acknowledge that this approach does not generate the exact cost of implementing a given recovery plan, both because the estimates are coarse, and because actual costs often vary from those estimated in the plan. However, our goal was to assign estimated costs to each plan that varied appropriately relative to other plans. For instance, expensive plans needed to be able to be considered appropriately in relation to less expensive plans. Our estimated costs for plan implementation fulfill this criteria and vary over seven orders of magnitude. All values were converted to 2016 US dollars ($) using the first recorded fiscal year as the year of recorded values, and consumer price index tables obtained from the Bureau of Labor Statistics (downloaded from https://download.bls.gov/pub/time.series/cu/cu.data.2.Summaries on 5/1/17). For all cost efficiency analyses, cost values were discounted into the future at a rate 3.7% based on the 50-year average annual increase in consumer price index.

Discussion of Figure

The figure represents an example of the benefit of optimal allocation. For an annual budget of $150 million, inefficient allocation of resources would recover about 104 species, while an optimal allocation is predicted to recover 1168 species. When annual funding is doubled to $300 million, the number of species recovered under the optimal allocation strategy increases to 1242.

The approach used to generate the data plotted in the "optimal allocation" curve is described in the preceding section. To obtain the "optimal allocation" curve we plotted the accumulation of species that could be recovered if spending on recovery plans was implemented in order of our calculated efficiency score ("E" in Equation 1) on the y axis. The x axis represents the cumulative amount that would be spent by all partners (USFWS, states, NGOs) to implement the plans in this order according to the estimated total cost of the plan (C) as described above.

The "inefficient allocation" curve is the mean and 90% prediction intervals of 100 random orders of recovery plan implementation using the same dataset as above. We used random selection of plans as a counterfactual for optimal funding in this figure because it represents a case where the various hidden objectives that go into the real-world decision making are unrelated to the value on the y-axis. The two curves converge at 1.7 billion, which represents the estimated total annual cost for actions needed to recover all endangered species in the data set. This is because the "inefficient allocation" approach is equivalent to the "optimal allocation" approach when all actions are funded.

The data and code for our figure are available at https://gitlab.sesync.org/ESA-decision-making/ESAFigure and data and code for the tool are available at https://gitlab.sesync.org/ESA-decision-making/shiny.

Supplementary Tables

Table S1. Benefit values for full funding of a recovery plan as a function of recovery potential and degree of threat.

		Degree of Threat (from RPN)		
		Low	Medium	High
Recovery Potential (from RPN)	Low	0.01	0.16	0.49
	High	0.33	0.66	0.99

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